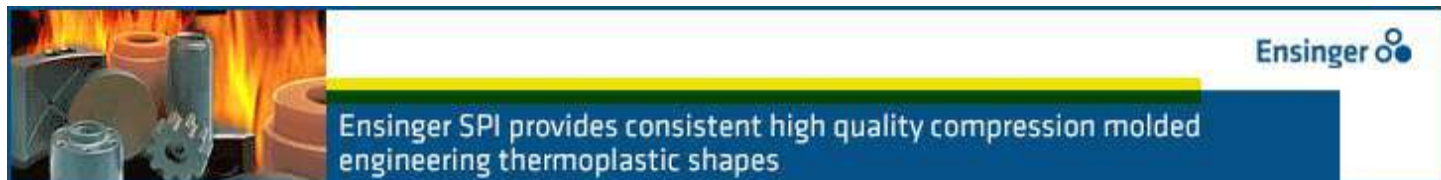



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Overview of materials for Polyamide-Imide, Molded

Categories: [Polymer](#); [Thermoplastic](#); [Polyamide-imide \(PAI\)](#); [Polyamide-Imide, Molded](#)

Material Notes: This property data is a summary of similar materials in the MatWeb database for the category "Polyamide-Imide, Molded". Each property range of values reported is minimum and maximum values of appropriate MatWeb entries. The comments report the average value, and number of data points used to calculate the average. The values are not necessarily typical of any specific grade, especially less common values and those that can be most affected by additives or processing methods.

Vendors: [Ensinger High Performance Plastics](#) withstand severe environments, resist harsh chemicals, and perform well in extreme temperatures. They offer weight savings that have potential to drive lower cost systems.

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Physical Properties	Metric	English	Comments
Density	1.40 - 1.58 g/cc	0.0506 - 0.0571 lb/in ³	Average value: 1.47 g/cc Grade Count:7
Water Absorption at Saturation	1.50 - 4.60 %	1.50 - 4.60 %	Average value: 2.20 % Grade Count:5
Mechanical Properties	Metric	English	Comments
Hardness, Rockwell E	70.0 - 80.0	70.0 - 80.0	Average value: 74.3 Grade Count:3
Hardness, Shore D	86.0 - 90.0	86.0 - 90.0	Average value: 88.5 Grade Count:4
Tensile Strength, Ultimate	83.0 - 150 MPa	12000 - 21800 psi	Average value: 116 MPa Grade Count:6
Tensile Strength	53.0 - 53.0 MPa @Temperature 235 - 235 °C	7690 - 7690 psi @Temperature 455 - 455 °F	Average value: 53.0 MPa Grade Count:1
Elongation at Break	2.00 - 6.40 %	2.00 - 6.40 %	Average value: 4.40 % Grade Count:3
Flexural Yield Strength	105 - 178 MPa	15200 - 25800 psi	Average value: 146 MPa Grade Count:6
Flexural Modulus	71.0 - 71.0 MPa @Temperature 235 - 235 °C	10300 - 10300 psi @Temperature 455 - 455 °F	Average value: 71.0 MPa Grade Count:1
Flexural Modulus	3.91 - 5.66 GPa	567 - 821 ksi	Average value: 4.59 GPa Grade Count:3
Compressive Yield Strength	31.0 - 172 MPa	4500 - 25000 psi	Average value: 90.0 MPa Grade C :5

	64.0 - 64.0 MPa @Temperature 235 - 235 °C	9280 - 9280 psi @Temperature 455 - 455 °F	Average value: 64.0 MPa Grade Count:1
Charpy Impact, Notched	0.400 - 1.50 J/cm ²	1.90 - 7.14 ft-lb/in ²	Average value: 0.767 J/cm ² Grade Count:3

Electrical Properties	Metric	English	Comments
Surface Resistance	1.00e+10 - 1.00e+14 ohm	1.00e+10 - 1.00e+14 ohm	Average value: 6.21e+13 ohm Grade Count:5
Dielectric Constant	3.34 - 6.00	3.34 - 6.00	Average value: 4.54 Grade Count:5
Dielectric Strength	13.7 - 24.0 kV/mm	348 - 610 kV/in	Average value: 19.5 kV/mm Grade Count:4
Dissipation Factor	0.00500 - 0.182	0.00500 - 0.182	Average value: 0.0411 Grade Count:5

Thermal Properties	Metric	English	Comments
CTE, linear	27.0 - 50.4 µm/m-°C	15.0 - 28.0 µin/in-°F	Average value: 37.6 µm/m-°C Grade Count:6
	40.0 - 50.0 µm/m-°C @Temperature >=150 °C	22.2 - 27.8 µin/in-°F @Temperature >=302 °F	Average value: 45.0 µm/m-°C Grade Count:2
Thermal Conductivity	0.259 - 0.540 W/m-K	1.80 - 3.75 BTU-in/hr-ft ² -°F	Average value: 0.385 W/m-K Grade Count:4
Maximum Service Temperature, Air	260 °C	500 °F	Average value: 260 °C Grade Count:5
Deflection Temperature at 1.8 MPa (264 psi)	257 - 279 °C	494 - 534 °F	Average value: 273 °C Grade Count:5
Minimum Service Temperature, Air	-50.0 - -20.0 °C	-58.0 - -4.00 °F	Average value: -30.0 °C Grade Count:3
Glass Transition Temp, Tg	275 - 280 °C	527 - 536 °F	Average value: 276 °C Grade Count:6

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.

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